Discrete Mathematics

Master of Computer Applications (MCA) — Semester 1 Complete Digital Study Notes

Course Code: MCA-101 / MCA-102

Format: Digital Notes & Examples

UNIT 1: SET THEORY, RELATIONS & MATHEMATICAL LOGIC

1. Sets & Relations

A **Set** is a collection of distinct, well-defined objects. A **Relation** R from set A to set B is a subset of the Cartesian product $A \times B$.

Properties of Binary Relations on a Set A:

- **Reflexive** $\forall a \in A, (a, a) \in R$
- **Symmetric** If $(a, b) \in R \implies (b, a) \in R$
- **Transitive** If $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$
- **Antisymmetric** If $(a, b) \in R$ and $(b, a) \in R \implies a = b$

2. Posets & Hasse Diagrams

A relation R on set A is a **Partially Ordered Set (POSET)** if it is Reflexive, Antisymmetric, and Transitive. It is denoted as $(A, \leq)$.

EXAMPLE: HASSE DIAGRAM CONSTRUCTION

Problem: Draw the Hasse Diagram for the poset $(D_{12}, |)$, where $D_{12} = \{1, 2, 3, 4, 6, 12\}$ and relation is divisibility ($a \mid b$).

Solution / Diagram Structure:

```
                [ 12 ]
               /                 [ 4 ]    [ 6 ]
              |     /                [ 2 ]    [ 3 ]
              \       /
                [ 1 ]
```

Note: In a Hasse diagram, reflexive loops and transitive edges are omitted, and directions move strictly upward.

3. Propositional Logic & Truth Tables

A proposition is a declarative statement that is either True (T) or False (F).

P	Q	P ∧ Q (AND)	P ∨ Q (OR)	P → Q (Implication)	P ↔ Q (Biconditional)
T	T	T	T	T	T

P	Q	P ∧ Q (AND)	P ∨ Q (OR)	P → Q (Implication)	P ↔ Q (Biconditional)
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

UNIT 2: ALGEBRAIC STRUCTURES, LATTICES & BOOLEAN ALGEBRA

1. Algebraic Structures Hierarchy

An algebraic structure consists of a non-empty set equipped with one or more binary operations satisfying specific axioms:

```
Set + Binary Operation (*)
    │
    ├──> [Associative] ──────────────────────> Semi-Group
    │                                              │
    │                                         + Identity (e)
    │                                              ▼
    │                                            Monoid
    │                                              │
    │                                          + Inverse (a⁻¹)
    │                                              ▼
    └────────────────────────────────────────> Group
                                                   │
                                               + Commutative
                                                   ▼
                                             Abelian Group
```

2. Lattices & Boolean Algebra

A lattice is a POSET $(L, \leq)$ in which every pair of elements $a, b \in L$ has a unique **Least Upper Bound (LUB** / **Join** ∨**)** and **Greatest Lower Bound (GLB** / **Meet** ∧**)**.

EXAMPLE: K-MAP MINIMIZATION

Problem: Simplify the Boolean function $F(A, B) = A'B + AB + AB'$ using a 2-variable Karnaugh Map.

```
          B = 0     B = 1
        +---------+---------+
  A = 0 |    0    |    1    |  <-- A'B
        +---------+---------+
  A = 1 |    1    |    1    |  <-- AB', AB
        +---------+---------+
```

Grouping:

- Group 1 (Row $A = 1$): Covers $AB' + AB = A$
- Group 2 (Column $B = 1$): Covers $A'B + AB = B$

Minimal Expression: $F(A, B) = A + B$

UNIT 3: COMBINATORICS & RECURRENCE RELATIONS

1. Pigeonhole Principle

If $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

EXAMPLE PROBLEM

Question: How many people must be in a room to guarantee that at least two share the same birth month?

Solution: Here, number of boxes (months) $k = 12$. By Pigeonhole Principle, we need $n = 12 + 1 = 13$ people.

2. Recurrence Relations

A recurrence relation defines a sequence a_n using previous terms.

SOLVED STEP-BY-STEP PROBLEM

Solve: $a_n - 5a_{n-1} + 6a_{n-2} = 0$ given initial conditions $a_0 = 1, a_1 = 4$.

Step 1: Characteristic Equation

Replace a_n with r^n:

$r^2 - 5r + 6 = 0 \Longrightarrow (r - 2)(r - 3) = 0$

Roots: $r_1 = 2, r_2 = 3$ (Real and Distinct)

Step 2: General Solution Form

$a_n = C_1(2)^n + C_2(3)^n$

Step 3: Apply Boundary Conditions

For $n = 0$: $C_1 + C_2 = 1$

For $n = 1$: $2C_1 + 3C_2 = 4$

Solving equations gives $C_1 = -1$ and $C_2 = 2$.

Final Solution:

$a_n = -(2^n) + 2(3^n)$

UNIT 4: GRAPH THEORY & TREES

1. Fundamentals & Handshaking Lemma

A graph $G = (V, E)$ consists of vertices V and edges E.

> **Handshaking Lemma:** For any undirected graph G:
> $\sum deg(v) = 2\,|E|$ (Sum of degrees of all vertices equals twice the number of edges)

2. Graphs Visualizations

```
(A)────────(B)           (A)────────(B)
 |  \     /  |            |          |
 |   \   /   |            |          |
 |     X     |            |          |
 |   /   \   |            |          |
 |  /     \  |            |          |
(C)────────(D)           (C)────────(D)

Complete Graph K₄ (6 Edges)   Planar Representation (No Edge Crossings)
```

3. Euler vs. Hamiltonian Graphs

- **Euler Path / Circuit:** Traverses every *edge* of the graph exactly once. (Exists if every vertex has an even degree).
- **Hamiltonian Path / Circuit:** Visits every *vertex* of the graph exactly once.